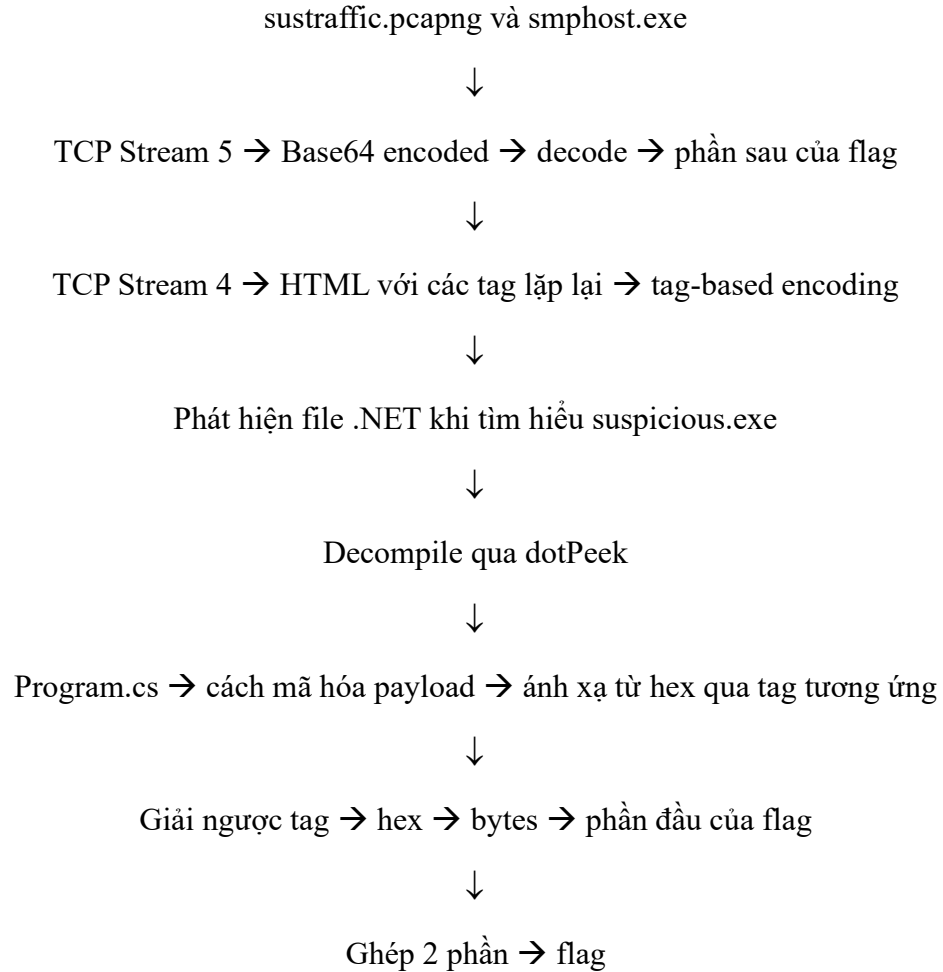


I. FISHY HTTP (EASY)

FLOW BÀI GIẢI:



- Đề cho 1 file .pcap và 1 file .exe, thông qua Wireshark phân tích file .pcap, follow TCP stream em thấy được tại stream 5



Với 2 dấu bằng ở cuối đoạn chữ nền xanh gần như chắc chắn là mã hóa Base64, và chương trình đang muốn ghép những chữ đầu tiên của mỗi từ để tạo thành chuỗi được mã hóa.

2. Ghép những chữ cái đầu của mỗi từ lại, ta có được chuỗi encode Base64, giải mã em có được phần thứ 2 của flag

```
C:\Users\ADMIN\Documents\Code\python>python a.py
1 string = input().split()
2 ans = ""
3 for c in string:
4     ans += c[0]
5 print(ans)

luongdung@DESKTOP-C ~$ ./mnt/c/Users/ADMIN/Downloads$ echo IFZvBHvtZSbpbIB
kcm12ZSBDIGhchcyBuby8sYmJlbc4NCiBwb2x1bWUgU2VyaWFsIE51bWJlcjBpcyBBMDc5LUFERKI
NCg0KIERpcmVjdG9yeSBvZiBDOlxUZW1wDQoNCjA1LzA3LzIwMjQIDA50jiyIEFNICAgIDxEsVI
+ICAgICAgICAgIC4NCjA1LzA3LzIwMjQIDA50jiyIEFNICAgIDxEsVI+ICAgICAgICAgIC4uDQo
wNS8wNy8yMDI0ICAwNzoyMyBBSAgICAgICAgNjcsNTE1LzA3LzIwMjQIDA50jiyIEFNICAgICAg
gICAgICAgICAgMSBgWxLkHMpICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg
yIERpcihzKSAgMjksNjM4LDUyMCw4MzIgYnVlLzA3LzIwMjQIDA50jiyIEFNICAgICAgICAgICAg
FTEx9JyANCg== | base64 -d
Volume in drive C has no label.
Volume Serial Number is A079-ADFB

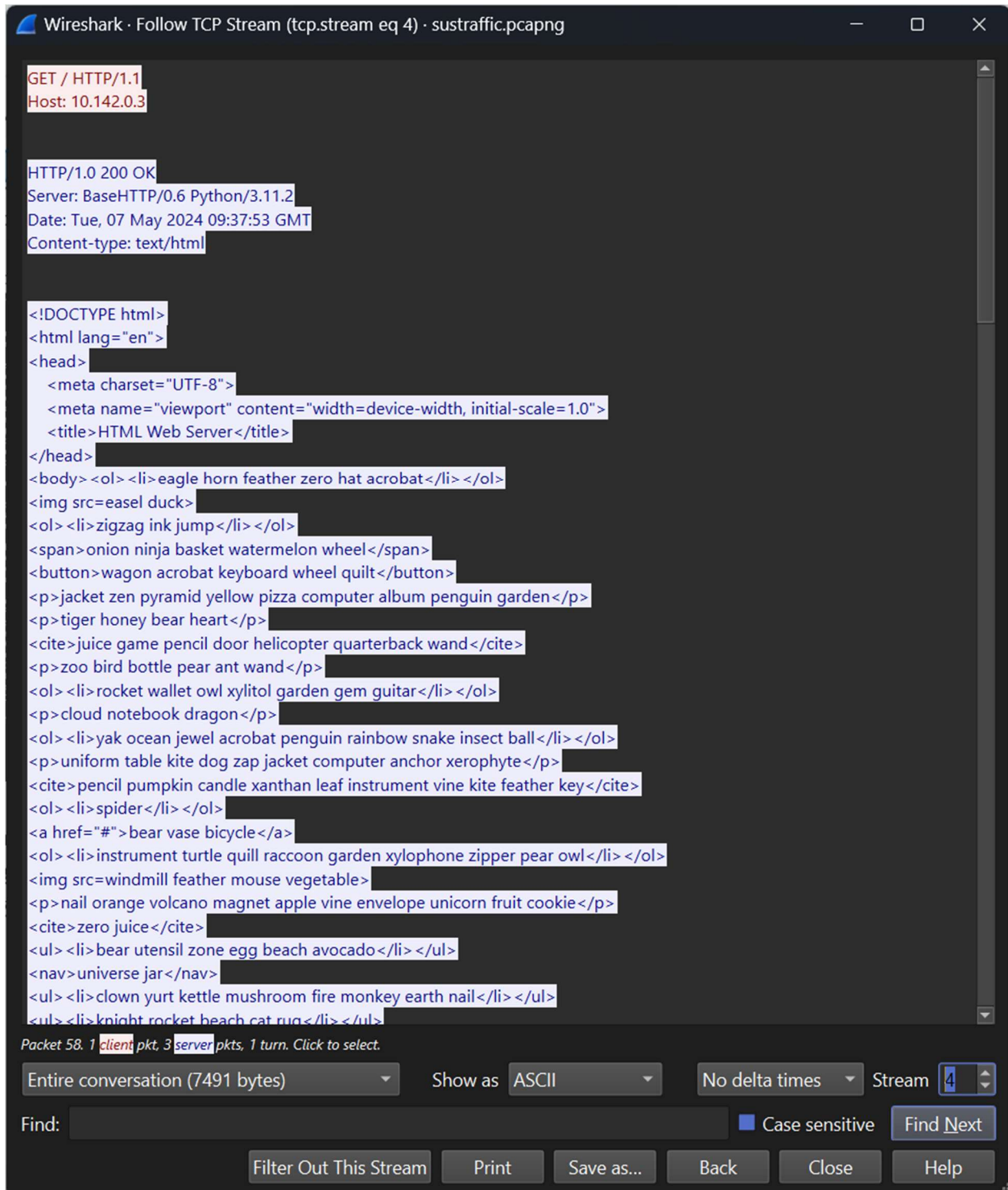
Directory of C:\Temp

05/07/2024 09:22 AM <DIR> .
05/07/2024 09:22 AM <DIR> ..
05/07/2024 07:23 AM 67,515,744 smphost.exe
1 File(s) 67,515,744 bytes
2 Dir(s) 29,638,520,832 bytes free
h77P_s73417hy_rev$HELL}
luongdung@DESKTOP-CBPS8LU:/mnt/c/Users/ADMIN/Downloads$ |

C:\Users\ADMIN\Documents\Code\python>python a.py
C:\Users\ADMIN\Documents\Code\python>python a.py
Invitation Fence Zombie vest book Horn Vegetable table Zipper Sun Bottle pencil but
terfly igloo Butterfly knight cloud moon log 2 Zebra Squirrel Basket Drum Iguana Gh
ost hat house cat yolk Ball universe beach yacht Bottle star Yellow Wing Jewel lady
bug ball Car 4 Necklace Castle instrument Ball Windmill basket 2 xanthan 1 bicycle
Wing Urn guitar Unicorn 2 Vacuum yolk astronaut Watermelon Flag snowman Igloo Escal
ator 5 1 ball Worm Jar laptop candle island Basket penguin cloud yogurt Butterfly B
icycle Magnet Drum candle 5 Ladybug Utensil Fish Engine Raccoon key Iron Ninja Comp
uter goat 0 Key Island Elephant Radio pear cat magnet Vase jet dragon Garden 9 yode
l eraser Snowman Basket van Zebra island Basket Drum Owl lamp xylophone Upholsterer
Zebra Whale 1 whale Dolphin Queen orange Nest Cat jet Ant 1 Log zone Apple 3 Light
house zelda Ice windmill Mountain jacket Quartz grape Ink Doll Astronaut 5 Olive ja
ck-o-lantern Iron yellow Instrument Elf Fence Net Ice Clown Ant game Invitation Dol
phin xerophyte Engine Snake Vest Ice + Igloo Candle Alarm gem Ink Candle Ant gift I
ce-cream Castle Album gem Iguana Cat 4 Nail Candle Jump Airplane 1 Lock zone Acroba
t 3 Laptop zigzag Instrument watch Monkey jacket Quartz glasses Iron Duck Airplane
5 Oyster jet Iron yak Ice Eraser Flower Newspaper Ice-cream Camera Airplane globe I
nk Dragon xmas Elephant Sandwich Vacuum Iron + Ink Castle Apple goat Igloo Cake App
le guitar Insect Camera Apple garden Igloo Clown 4 universe Dolphin Quill oven worm
Newspaper Squirrel 8 windmill Nail yacht 8 yacht Mouse Dinosaur Iguana 0 Igloo Cas
tle Anchor windmill Nest zap ornament yodel Mushroom yodel Basket Ball Turtle Sock
Anchor globe Ink Castle Anchor goat Iron Cake Alarm guitar Notebook juice clown soc
k Ninja Table Egg 1 Lighthouse Duck computer 0 Ninja Camera Banana zoo bear Xanthan
Beach oyster bear 3 Nail 0 Laptop mountain Vegetable 4 Zen Quarterback 0 Hoala Ice
Cat Acrobat game Insect Cat Apple gift Igloo Clown Alarm gift Iron Cookie Acrobat
gem Instrument Camera Astronaut guitar Mountain Ship Beach Globe album Wand x-ray l
amp Kangaroo House Mailbox penguin Invitation Castle Ant grape Igloo Cat Ant 2 Ninj
a yawn whale 1 Microphone Table Unicorn sock Net zipper Quiver 0 Iron Game Juice 5
diamond Gem Violin zen Door Quadrilateral oyster ghost Igloo Cat Apple guitar Ink C
amera Acrobat garden Ice Cake Avocado grape Iguana Cat Acrobat game Ink Computer Av
ocado yogurt Ink Easel Rose popcorn castle ice-cream heart zap Kitchen Star Astrona
ut globe Magnet jet key sock Notebook jewel Mailbox 4 Log Desk Unicorn yogurt Mask
Cookie watch 4 Magnet zebra Ice garden Yurt necklace leaf 0 Zap Xylophone Mask ghos
t Zelda newspaper Jet Lighthouse Zipper Quail 0 Kitchen Jigsaw 2 game 3 Necklace 1
Beach fire candle zombie camera zoo Newspaper Diamond Eraser 3 astronaut Horn ladyb
ug fence camera mailbox Van 2 Uniform 0 hamburger Fox Trumpet Eagle xylylot 9 Jack-
o-lantern yolk Astronaut Net candle guitar = =
IFZvBHvtZSbpbIBkcm12ZSBDIGhchcyBuby8sYmJlbc4NCiBwb2x1bWUgU2VyaWFsIE51bWJlcjBpcyBBMDc5LUFERKI
NCg0KIERpcmVjdG9yeSBvZiBDOlxUZW1wDQoNCjA1LzA3LzIwMjQIDA50jiyIEFNICAgIDxEsVI+ICAgICAgICAgIC4NCjA1LzA3LzIwMjQIDA50jiyIEFNICAgIDxEsVI+ICAgICAgICAgIC4uDQo
wNS8wNy8yMDI0ICAwNzoyMyBBSAgICAgICAgNjcsNTE1LzA3LzIwMjQIDA50jiyIEFNICAgICAggICAgICAgICAgMSBgWxLkHMpICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAgICAg
yIERpcihzKSAgMjksNjM4LDUyMCw4MzIgYnVlLzA3LzIwMjQIDA50jiyIEFNICAgICAgICAgICAgFTEx9JyANCg==
FTEx9JyANCg==

C:\Users\ADMIN\Documents\Code\python>
```


3. Quay lại với stream 4, em thấy một đoạn mã HTML đơn giản, với những thẻ được lặp lại như `<button>`, `<ol>`, `<p>` tuy rằng nội dung bên trong thẻ thì hoàn toàn vô nghĩa. Có thể attacker đã sử dụng bảng mã bí mật để truyền thông tin qua những thẻ tags này.



The image shows a Wireshark packet capture window titled "Follow TCP Stream (tcp.stream eq 4) · sustraffic.pcapng". The packet list on the left shows "GET / HTTP/1.1" and "Host: 10.142.0.3". The packet details pane shows the HTTP response structure: "HTTP/1.0 200 OK", "Server: BaseHTTP/0.6 Python/3.11.2", "Date: Tue, 07 May 2024 09:37:53 GMT", and "Content-type: text/html". The packet bytes pane displays the HTML content, which is a list of items organized into various HTML tags like `<ol>`, `<p>`, `<img>`, `<span>`, `<button>`, `<cite>`, `<a href="#">`, `<ul>`, and `<nav>`. The items are listed in a seemingly random order, such as "eagle horn feather zero hat acrobat", "onion ninja basket watermelon wheel", "wagon acrobat keyboard wheel quilt", "jacket zen pyramid yellow pizza computer album penguin garden", "tiger honey bear heart", "juice game pencil door helicopter quarterback wand", "zoo bird bottle pear ant wand", "rocket wallet owl xylitol garden gem guitar", "cloud notebook dragon", "yak ocean jewel acrobat penguin rainbow snake insect ball", "uniform table kite dog zap jacket computer anchor xerophyte", "pencil pumpkin candle xanthan leaf instrument vine kite feather key", "spider", "bear vase bicycle", "instrument turtle quill raccoon garden xylophone zipper pear owl", "windmill feather mouse vegetable", "nail orange volcano magnet apple vine envelope unicorn fruit cookie", "zero juice", "bear utensil zone egg beach avocado", "universe jar", "clown yurt kettle mushroom fire monkey earth nail", and "knight rocket beach cat rug". The packet list at the bottom shows "Packet 58. 1 client pkt, 3 server pkts, 1 turn. Click to select." The packet details pane shows "Entire conversation (7491 bytes)" and "Show as ASCII". The packet bytes pane shows "No delta times" and "Stream 4". The packet list pane shows "Find:" and "Find Next".

```
GET / HTTP/1.1
Host: 10.142.0.3

HTTP/1.0 200 OK
Server: BaseHTTP/0.6 Python/3.11.2
Date: Tue, 07 May 2024 09:37:53 GMT
Content-type: text/html

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Web Server</title>
</head>
<body>
<ol>
<li>eagle horn feather zero hat acrobat</li>
</ol>
<img src=easel duck>
<ol>
<li>zigzag ink jump</li>
</ol>
<span>onion ninja basket watermelon wheel</span>
<button>wagon acrobat keyboard wheel quilt</button>
<p>jacket zen pyramid yellow pizza computer album penguin garden</p>
<p>tiger honey bear heart</p>
<cite>juice game pencil door helicopter quarterback wand</cite>
<p>zoo bird bottle pear ant wand</p>
<ol>
<li>rocket wallet owl xylitol garden gem guitar</li>
</ol>
<p>cloud notebook dragon</p>
<ol>
<li>yak ocean jewel acrobat penguin rainbow snake insect ball</li>
</ol>
<p>uniform table kite dog zap jacket computer anchor xerophyte</p>
<cite>pencil pumpkin candle xanthan leaf instrument vine kite feather key</cite>
<ol>
<li>spider</li>
</ol>
<a href="#">bear vase bicycle</a>
<ol>
<li>instrument turtle quill raccoon garden xylophone zipper pear owl</li>
</ol>
<img src=windmill feather mouse vegetable>
<p>nail orange volcano magnet apple vine envelope unicorn fruit cookie</p>
<cite>zero juice</cite>
<ul>
<li>bear utensil zone egg beach avocado</li>
</ul>
<nav>universe jar</nav>
<ul>
<li>clown yurt kettle mushroom fire monkey earth nail</li>
</ul>
<ul>
<li>knight rocket beach cat rug</li>
</ul>

```

4. Phân tích qua file .exe với lệnh strings, em nhận thấy rằng đây có thể là một file .NET

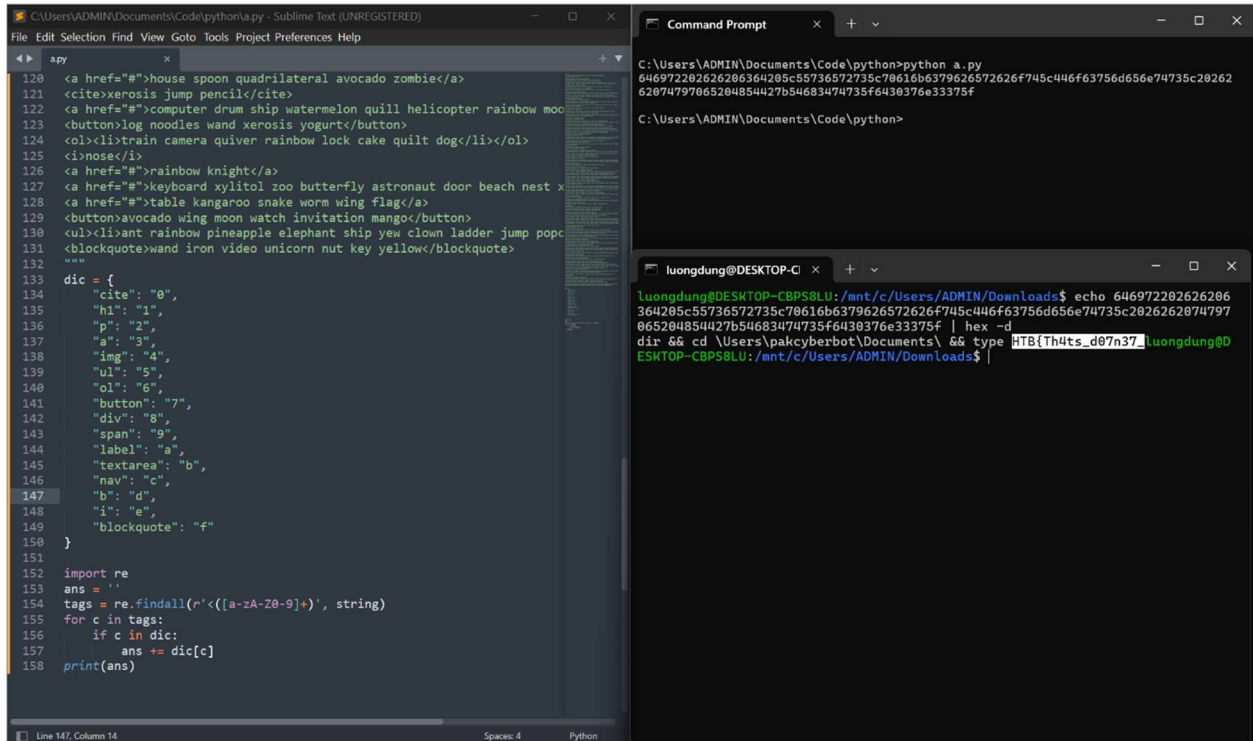
```
luongdung@DESKTOP-CL x + v - □ ×
System.Net.Security
XmlEntity
IIdentity
GenericIdentity
HttpListenerBasicIdentity
ClaimsIdentity
System.Net.WebProxy
IWebProxy
AssemblyNameProxy
System.Reflection.DispatchProxy
}*:S'
RSA1
WrapNonExceptionThrows
.NETCoreApp,Version=v8.0
FrameworkDisplayName
.NET 8.0
Serviceable
True
PreferInbox
True
netstandard
IsTrimmable
True
Microsoft Corporation
Microsoft Corporation. All rights reserved.
8.0.424.16909
.8.0.4+2d7eea252964e69be94cb9c847b371b23e4dd470
Microsoft
.NET
RepositoryUrl!https://github.com/dotnet/runtime
```

5. Dùng dotPeek và tìm file Program, em đã thấy được mã nguồn viết theo ngôn ngữ C lộ ra quy trình mà attacker đã mã hóa mã độc của mình thông qua những tags ở trên, cách mã hóa thông qua

```
static string DecodeData(string data)
{
    StringBuilder stringBuilder = new StringBuilder();
    foreach (Match match in new Regex("<\\w+\\s>", RegexOptions.IgnoreCase).Matches(((Capture) new Regex("<body>.*?</body>", RegexOptions.IgnoreCase).Match(data)
    {
        Environment.NewLine
    }, (StringSplitOptions) 1)[0]))
    {
        if (((Group) match).Success)
        {
            GroupCollection groups = match.Groups;
            if (string.op_Inequality(((Capture) groups[1]).Value, "li"))
                stringBuilder.Append(Program.tagHex[(((Capture) groups[1]).Value)]);
        }
    }
    return HexStringToBytes(stringBuilder.ToString());
}

static string HexStringToBytes(string hex)
{
    byte[] numArray = new byte[hex.Length / 2];
    for (int index = 0; index < hex.Length; index += 2)
        numArray[index / 2] = Convert.ToByte(hex.Substring(index, 2), 16);
    return Encoding.ASCII.GetString(numArray);
}
```

6. Ngoài ra em cũng thấy dictionary cho quá trình giải mã ở cùng file này, attacker đã chuyển chuỗi mã độc theo dạng hex, và tương ứng 15 tags sẽ ứng với 0,1,2...,e,f. Từ đó em giải ra đoạn mã hex ban đầu và tìm được phần đầu của flag.



The screenshot shows a Sublime Text editor window on the left and a Windows Command Prompt window on the right. The Sublime Text window is open to a file named 'a.py' and displays a Python script. The script defines a dictionary 'dic' mapping HTML tags to characters (0-9, a-f) and uses a regular expression to find these tags in the input string. The Command Prompt window shows the execution of the script, which outputs a long hexadecimal string. Below this, a directory listing is shown, and a file named 'HTB{Th4ts_d07n37_lu0ngdung@DESKTOP-CBPS8LU: /mnt/c/Users/ADMIN/Downloads\$ |' is highlighted.

```
C:\Users\ADMIN\Documents\Code\python>python a.py
6469722026206364205c55736572735c70616b6379626572626f745c446f63756d656e74735c20262
62074797065204854427b54683474735f6430376e33375f

C:\Users\ADMIN\Documents\Code\python>

luongdung@DESKTOP-CBPS8LU: /mnt/c/Users/ADMIN/Downloads$ echo 6469722026206
364205c55736572735c70616b6379626572626f745c446f63756d656e74735c20262074797
065204854427b54683474735f6430376e33375f | hex -d
dir && cd \Users\pakcyberbot\Documents\ && type HTB{Th4ts_d07n37_lu0ngdung@DE
SKTOP-CBPS8LU: /mnt/c/Users/ADMIN/Downloads$ |
```

Ghép 2 phần của flag tìm được em có được flag của bài.